

Computational optimization of a vehicle dynamic simulation model

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Abstract—The importance of simulation in motorsport racing has steadily increased. Fine-tuning these simulators is complex and traditionally manual, relying on engineers' expertise.

This study develops a virtual driver and automates a parameter tuning pipeline for a race car simulator. The virtual driver based on the Data-enabled Predictive Control (DeePC) algorithm outperforms traditional Proportional Integral Derivative (PID) and Model Predictive Control (MPC) controllers and can potentially be extended to compute an optimal trajectory.

A YAML-configured optimization pipeline with a custom Domain-Specific Language (DSL) was developed to accelerate parameter tuning. However, parameter interdependency is problematic for traditional optimization algorithms. Bayesian optimization can potentially find a global minimum despite interdependencies and converge to the optimum faster than traditional optimization methods.

Index Terms—Racecar Simulation, Virtual Driver, Parameter Tuning,

I. INTRODUCTION

The importance of simulation in motorsport racing has been steadily increasing over the years, with teams investing millions of dollars to gain an edge over their competitors. Full motion racing simulators are widely used not only for driver training but also for design purposes, thanks to their many advantages over real track training. A better simulation setup and a high correlation between the simulator and the track can lead to significant gains in cost savings, accuracy, and time efficiency.

Fine-tuning the simulator parameters is a complex task, especially with the growing number of variables involved. Currently, this tuning process is manual [1] and in the early stages of development, relying heavily on engineers' experience and the existent theoretical frameworks to guide it. Although manual tuning is time-consuming, it is still commonly used.

For the simulation to provide valuable feedback, it must be driven by a professional Driver-in-Loop (DiL) [2]. This

This work is supported by the project "Ottimizzazione computazionale di un modello di simulazione dinamica del veicolo" funded by the European Union - NextGenerationEU and Dallara Automobili S.p.A. through the Italian "National Recovery and Resilience Plan" (PNRR) Mission 4, Component 2, Investment 3.3 (DM 352/2022) - CUP J33C22001400009

requirement limits the use of DiL simulators because a professional driver must be physically present. Alternatively, accurately simulating a driver's behavior (similar to programming virtually driven cars) is crucial for obtaining accurate feedback faster.

The work was divided into two parts:

- creating a virtual driver that can accurately simulate the behavior of a professional driver; and
- developing an automated system that can streamline and speed up the parameter tuning process.

II. VIRTUAL DRIVER

The virtual driver building process can be divided as:

- compute the optimal trajectory based on the race track limits and car dynamics;
- compute the control actions based on pre-computed optimal trajectory.

The first sub-problem can be solved by using a trajectory from real data. It respects the track limits but does not completely match the car dynamics.

The second sub-problem can be solved by using industrial first-choice PID controller [3]. It is a simple and effective solution, but it is not optimal. Academic state-of-the-art MPC controller [4] can outperform PID controller, but it is computationally expensive and requires an explicit model of the car.

It was proposed to use a DeePC algorithm [5]. It has been found that the approach is viable, and can outperform traditional controllers PID and MPC, especially on the verge of the car's performance limits. Additionally, its data-driven nature makes it suitable for problems with computationally expensive high-end models of the race car.

Moreover, a DeePC algorithm can be extended to compute the optimal trajectory. The optimal trajectory used in DeePC algorithm is substituted by racetrack limits [6] and the cost function is modified to penalize low speed. The algorithm is forced to find such trajectory that fits into the track limits and has the highest average speed along the track.

However, optimal trajectory generation requires an extended prediction horizon that covers at least the next corner [6]. It may cause computational complexity issues [7].

III. PARAMETER TUNING

Parameter tuning of the simulator is an optimization problem to find the best set of parameters that minimize the error between the simulated data and the real car data in the same conditions. Directly comparing the real car data with simulated data does not yield meaningful results due to a high degree of uncertainty and variable interdependency.

The high degree of uncertainty in manual tuning, is mitigated by numerous metrics introduced in the relevant literature [8], [9], [10]. The key idea of these metrics is to align pre-processed data from the real track and simulation through a scatter plot. Manual alignment is so common that proprietary data analysis software, such as MoTec i2 Pro¹ is widely used.

To automate alignment, the optimization pipeline was developed. It uses a cost function based on those metrics. Metrics of the optimization pipeline are specified in YAML. To support the user, a custom DSL is used to set up the optimization configuration. The use of DSL makes configuration of the pipeline easier for vehicle engineers. It includes access to the data, data filters, mathematical channels, scatter plots configuration, matrix operations, trigonometrical operations, simple cycles etc.

The optimization pipeline not only accelerates the process of parameter tuning but also tackles the problem with interdependency of variables. In the manual tuning process, interdependent variables are tuned iteratively one by one, while an automated pipeline can define multi-dimensional cost function to optimize multiple variables simultaneously.

Interdependent variables are often encountered in race car parameters such as tires, suspension, and aerodynamics. The cost function of interdependent variables can form a “narrow valley”, which poses a common challenge for optimization algorithms. Benchmarking of the most common optimization algorithms in Pytorch framework² shows that AdamP [11] and SWATS [12] outperform traditional optimization algorithms such as Adam [13], RMSprop [14].

An alternative optimization algorithm is Bayesian optimization. Frameworks such as AutoML [15], SMAC [16], and Optuna [17]. Unlike standard optimization algorithms, Bayesian optimization, due to its probabilistic nature [18], can find the global minimum in narrow valley problems. Additionally, Bayesian optimization can use multifidelity simulation [19] to optimize simulator parameters for specific track sections, narrowing the search space based on the initial track portion and potentially achieving faster convergence to the global minimum.

IV. CONCLUSION

The race car simulation is a complex task that presents numerous challenges, including virtual driver implementa-

tion and optimization of interdependent parameters.

To enhance the race car simulator, a virtual driver utilizing the DeePC algorithm was incorporated alongside an optimization pipeline. This improvement enables more precise tuning of simulator parameters. However, the challenge of generating an optimal trajectory remains.

Future work will focus on extending the DeePC algorithm to compute the optimal trajectory. Additionally, Bayesian optimization will be explored for parameter tuning, including multifidelity approaches.

REFERENCES

- [1] M. Guiggiani, *The Science of Vehicle Dynamics*. Springer Dordrecht, 2016.
- [2] G. Napolitano Dell’Annunziata, R. Maglione, A. Genovese, A. Sakhnevych, F. Timpone, and F. Farroni, “Sport driving skills: A preliminary comparative study from outdoor testing sessions,” *Transportation Research Interdisciplinary Perspectives*, vol. 25, p. 101105, May 2024. [Online]. Available: <http://dx.doi.org/10.1016/j.trip.2024.101105>
- [3] W. Farag, “Complex trajectory tracking using PID control for autonomous driving,” *International Journal of Intelligent Transportation Systems Research*, vol. 18, no. 2, pp. 356–366, 2019.
- [4] M. L. Darby and M. Nikolaou, “MPC: Current practice and challenges,” *Control Engineering Practice*, vol. 20, no. 4, pp. 328–342, 2012.
- [5] J. Coulson, J. Lygeros, and F. Dorfler, “Data-enabled predictive control: In the shallows of the deepc,” in *18th ECC*, 2019, pp. 307–312.
- [6] A. Liniger, A. Domahidi, and M. Morari, “Optimization-based autonomous racing of 1:43 scale RC cars,” *Optimal Control Applications and Methods*, vol. 36, no. 5, pp. 628–647, 2014.
- [7] D. Arnström, “Real-time certified MPC: reliable active-set QP solvers,” Ph.D. dissertation, Linköping University, Sweden, 2023.
- [8] B. Christopher, *Making Sense of Squiggly Lines : The Basic Analysis of Race Car Data Acquisition*. CB-Racing, 2011.
- [9] B. Knox, *A Practical Guide to Race Car Data Analysis*. CreateSpace Independent Publishing Platform, 2011.
- [10] S. Derek, *Race car design*. PALGRAVE, 2014.
- [11] B. Heo, S. Chun, S. J. Oh, D. Han, S. Yun, G. Kim, Y. Uh, and J. Ha, “Adamp: Slowing down the slowdown for momentum optimizers on scale-invariant weights,” in *ICLR*, 2021.
- [12] N. S. Keskar and R. Socher, “Improving generalization performance by switching from adam to SGD,” *CoRR*, vol. abs/1712.07628, 2017.
- [13] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *ICLR*, Y. Bengio and Y. LeCun, Eds., 2015.
- [14] A. Graves, “Generating sequences with recurrent neural networks,” *CoRR*, vol. abs/1308.0850, 2013.
- [15] H. Eldeeb, M. Maher, R. E. Shawi, and S. Sakr, “Automl-bench: A comprehensive experimental evaluation of automated machine learning frameworks,” *Expert Syst. Appl.*, vol. 243, p. 122877, 2024.
- [16] M. I. Fakhir, S. A. R. Kazmi, A. Qasim, and A. Ishaq, “SMACS: A framework for formal verification of complex adaptive systems,” *Open Comput. Sci.*, vol. 13, no. 1, 2023.
- [17] Z. Yan, Z. Qin, J. Fan, Y. Huang, Y. Wang, J. Zhang, L. Zhang, and Y. Cao, “Gas outburst warning method in driving faces: Enhanced methodology through optuna optimization, adaptive normalization, and transformer framework,” *Sensors*, vol. 24, no. 10, p. 3150, 2024.
- [18] S. Cheekaty and G. Muneeswari, “Exploring sparse gaussian processes for bayesian optimization in convolutional neural networks for autism classification,” *IEEE Access*, vol. 12, pp. 10 631–10 651, 2024.
- [19] X. Gong and Y. Pan, “Multifidelity bayesian experimental design to quantify rare-event statistics,” *SIAM/ASA J. Uncertain. Quantification*, vol. 12, no. 1, pp. 101–127, 2024.

¹<https://www.motec.com.au/i2/i2pro/>

²<https://pytorch.org>